Which Employers Alter Offers of Health Insurance In Response to Minimum Wage Increases? Evidence from the Medical Expenditure Panel Survey - Insurance Component

by

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Abstract

Offers of health insurance vary significantly across firm and establishment characteristics, including firm size, workforce sex, and workforce age (Agency for Healthcare Research and Quality, 2024, Table I.A.2; Abraham et al., 2016). Low-wage workers in particular are less likely to receive health insurance from their employer, and employers tend to roll back benefits as minimum wages rise (Meiselbach and Abraham, 2023; Marks, 2011). This study aims to identify the effects by employer and workforce characteristics of minimum wage increases on employer-provided benefits using restricted-use data from the Medical Expenditure Panel Survey-Insurance Component (MEPS-IC). I use a two-way fixed effects approach, exploiting state legislation variation, to identify changes in the rates of employers offering health insurance and employer and employee characteristics associated with changes in health insurance offers. I study employer and employee characteristics for which minimum wage responses may differ due to differences in health insurance costs, impacts of the Affordable Care Act and other policies and legislation, and employee preferences for health insurance. I find that establishments alter offers of health insurance in response to the minimum wage differently based on firm size and workforce gender and age. Small firms drive reductions in health insurance offers as the minimum wage increases, and reductions are smaller among workforces with a greater share of female or older workers, who more highly value employer-provided health insurance.

Keyword: minimum wage, employer health insurance

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Introduction

Minimum wage increases raise labor costs for employers, who may adjust labor practices to minimize cost increases. They can do so through a variety of adjustments, including but not limited to reducing employment, hours, benefits, or training (Clemens, 2021; Schmitt, 2015). Evidence of effects on employment indicate negative to zero effects on employment from minimum wage increases (Azar et al., 2024; Clemens and Wither, 2019; Schmitt, 2015; Neumark et al., 2014; Sabia et al., 2012; Allegretto et al., 2011; Dube et al., 2010). Employment effects may be close to zero if employers choose to adjust over other margins, such as benefits, but employee welfare may fall even if employment does not (Clemens, 2021).

With regard to voluntary benefits, there is evidence that employers adjust health insurance offers in response to rising labor costs. Health insurance is a costly benefit to provide, and an increase in the wage floor will increase the cost of the total compensation bundle, assuming benefits remain constant. Prior literature indicates that with a $1 increase in the minimum wage, the share of low-wage workers receiving health insurance from their employer falls approximately 1 percentage point (Dworsky et al., 2022; Clemens et al., 2018; Marks, 2011). Absent minimum wage changes, health insurance offers also vary significantly across firm, establishment, and workforce characteristics due to differences in benefit costs and policy – for example, in 2024, 32 percent of firms with 49 or fewer employees offered health insurance, compared to 96 percent of larger firms (Agency for Healthcare Research and Quality, 2024, Table I.A.2). Similarly, Clemens (2021) notes that the margins over which employers may adjust after a minimum wage increase – such as by raising prices or altering benefits – may vary by factors such as industry and workforce preferences. Thus, it is reasonable to predict that

minimum wage effects on health insurance offers may differ across a variety of business characteristics.

In this project, I use Medical Expenditure Panel Survey - Insurance Component (MEPS-IC) microdata to study effects of minimum wage increases on employer health benefit offers, controlling for and considering effects by workforce characteristics. The MEPS-IC provides detailed information on health benefits offered by employers across the nation, as well as employer characteristics. This dataset allows for employer-side analysis of changes in health insurance offers, whereas previous studies use employee-side household surveys to measure changes in employer health insurance receipt (Dworsky et al., 2022; Clemens et al., 2018; Marks, 2011; Simon and Kaestner, 2004). Employee-side data additionally captures effects on employee health benefit take-up and employment, while employer-side data isolates effects of the minimum wages on employer decisions. I expand upon the work of Meiselbach and Abraham (2023), who use MEPS-IC state-year summary statistics, to study additional sources of heterogeneity which are associated with differences in health benefit costs, policy, and employee preferences. Thus, this dataset allows for both analysis of the first-order question – which employers change their offers of health insurance when the minimum wage increases? – and suggests what factors are relevant in an employer's decision to alter health benefit offers when the wage floor rises.

I use data from 2003 to 2023, covering a period over which the federal minimum wage increased (between 2007 and 2009), and the Affordable Care Act (ACA) was implemented (2014) to examine this question. I use a two-way fixed effects approach, exploiting state legislation variation, to identify heterogeneous effects of the minimum wage on health insurance offers by firm size and the share of workers that are female, aged 50 or older, and high-wage.

The interaction of firm size and workforce characteristics identifies separate effects for establishments with differing valuation of and preferences for health insurance (workforce characteristics) and differing health care costs and relevant policies (firm size). Additionally, I identify effects before and after implementation of the ACA, which had specific impacts based on firm and establishment characteristics.

I find that offers of health insurance fall approximately 0.5 percentage points with a $1 increase in the minimum wage, driven by small firms (49 or fewer employees) which see a decline in health insurance offers of 0.6 percentage points. Declines in health insurance offers as the minimum wage rises are larger in the pre-ACA period. Both small-firm establishments with a large share of low-wage and high-wage employees see declines in health insurance offers, suggesting that health insurance costs, which are higher for smaller firms, are important in determining health insurance offer responses to minimum wage increases. Establishments in small firms with 66 percent or more female employees see smaller declines in health insurance offers as the minimum wage increases (and surprisingly, a 0.5 percentage point increase in the post-ACA period) than small-firm establishments with fewer female employees, consistent with women more highly valuing employer-provided health insurance than men, employers considering employee preferences for benefits, and changes in premium rating based on sex in the small-group health insurance market after the ACA. Small-firm establishments with less than 20 percent of workers aged 50 or older see larger reductions in health insurance offers from a minimum wage increase than among establishments with 50 percent or more older workers, with older workers also more highly valuing health insurance offers from their employer. Establishments in medium and large firms (50-499 employees and 500 or more employees, respectively) largely do not change health insurance offers in response to minimum wage

increases, consistent with these firms being more likely to offer health insurance and provisions of the ACA which encourage them to provide health insurance.

This paper studies one area of compensation that may be affected by minimum wage increases and considers heterogeneous minimum wage effects which are previously unexplored. While employment effects of the minimum wage have been extensively studied (Azar et al., 2024; Neumark and Shirley, 2022; Clemens and Wither, 2019; Schmitt, 2015; Neumark et al., 2014; Sabia et al., 2012; Allegretto et al., 2011; Dube et al., 2010; Sabia, 2009; Card and Krueger, 1994), effects on the total compensation bundle, including non-wage compensation, have more minimal evidence. Since provision of health insurance depends on employer costs, employee preferences, and policy mandates, responses to minimum wage increases likely vary across firm and establishment characteristics, indicating that there is more variation than previously documented (Meiselbach and Abraham, 2023; Dworsky et al., 2022; Clemens et al., 2018; Marks, 2011). I find evidence that offers of health insurance are more responsive to minimum wage changes in some firms and establishments, and document what firm and establishment characteristics are associated with declines in employer-provided health insurance offers.

Background

Minimum Wage on Employer Health Insurance Offers

Minimum wage increases cause employment costs to increase through wages, and one option employers may choose to smooth labor cost changes (in addition to or alternatively of reducing employment or hours) is to tradeoff wages and benefits. Minimum wages set a mandated wage floor, and thus employers may reduce non-mandated forms of compensation

such as health insurance to minimize changes in the compensation bundle (Clemens, 2021; Lee and Warren, 1999). Under compensating wage differentials, employees with health insurance receive lower wages than those without, and the imposition of wage floors means that when wages cannot be reduced, employer-provided health insurance may be eliminated (Qin and Chernew, 2014; Sherstyuk et al., 2014; Lluis and Abraham, 2013; Olson, 2002).

Consumer-side surveys confirm that employer-provided health insurance enrollment falls for low-wage or low-skilled workers as the minimum wage rises (Dworsky et al., 2022; Lenhart, 2020; Clemens et al., 2018; Marks, 2011; Royalty, 2000). These studies reflect effects that may arise from changes in employer health insurance offers – such as choosing to decline to offer it in response to rising labor costs – and/or changes in employee take-up – for example, employers may change features of the offered health insurance so that it is less affordable or desirable for employees. Marks (2011) finds that low-skilled workers in smaller firms are less likely to receive employer-provided health insurance and less likely to be enrolled in a plan with no employee premium contribution required. Dworsky et al. (2022) and Clemens et al. (2018) both find that the decline in employer-provided health insurance enrollment remains conditional on employment, such that minimum wages affect benefit receipt even in the absence of disemployment effects.

Employer-side surveys such as the MEPS-IC allow for analysis of the first-order response of businesses to minimum wage changes.[1] Meiselbach and Abraham (2023) use the MEPS-IC summary statistics and find that a $1 increase in the minimum wage reduces the share of employers offering health insurance 0.9 percentage points, with larger effects among small firms

[1] There are also issues of reliability in studying health insurance sources in consumer surveys. Davern et al. (2008) and Nelson et al. (2000) find that survey respondents are likely to correctly report that they have health insurance but frequently misreport their source of health insurance. Employers are presumably more likely to correctly report if their business offers health insurance.

and employers with a large share of low-wage workers. I expand upon Meiselbach and Abraham (2023) by using MEPS-IC microdata to control for and identify heterogeneity by additional establishment characteristics.

Heterogeneity in Health Insurance Offers

Even in the absence of minimum wage changes, offers of employer-provided health insurance vary significantly by firm, establishment, and workforce characteristics, including firm size, wage levels, worker sex, and worker age. In 2024, 37 percent of businesses with a majority of low-wage workers offered health insurance, compared to 53 percent of businesses with fewer than 50 percent low-wage workers (Agency for Healthcare Research and Quality, 2024, Table I.A.2).[2] Low-wage employers are consistently less likely to offer health insurance, and low-wage workers are less likely to have access to health insurance from their employer (Long and Marquis, 2001; Marquis and Long, 2001).

Offers of health insurance vary even more by number of employees. In 2024, 96 percent of firms with 50 or more employees offered health insurance, compared to 32 percent of firms with fewer than 50 employees (Agency for Healthcare Research and Quality, 2024, Table I.A.2). Firm size also determines impacts of the ACA's implementation in 2014. Notably, the ACA places requirements on firms with 50 or more full-time employees to offer minimum health insurance coverage or pay an employer shared responsibility payment.[3] Larger firms tend to self-insure health benefits, paying directly for medical claims rather than purchasing insurance coverage, which is cheaper for larger firms and requires compliance with federal insurance laws,

[2] "Low-wage" is defined as a worker receiving wages below a nominal dollar value stated in the Medical Expenditure Survey questionnaire. In 2003, this was $9.50 per hour, and in 2023, this was $15.50 per hour. Table 1 lists this value in each survey year from 2003 to 2023.

[3] See https://www.irs.gov/affordable-care-act/employers/affordable-care-act-tax-provisions-for-large-employers and https://www.irs.gov/affordable-care-act/employers/employer-shared-responsibility-provisions.

not state insurance laws (Fronstin, 2016; Buettgens and Blumberg, 2012).[4] Important among those federal laws is a provision that businesses which self-insure their health benefit plans are bound by non-discrimination rules which limit offering different plans to workers based on compensation.

If there is a fixed cost of searching for and administering insurance, larger firms are more likely to offer health insurance because the average cost per worker declines as the number of employees and employee hours increase (Hadley and Reschovsky, 2002). Smaller firms face smaller risk pools (and higher premiums, conditional on equivalent benefits), particularly before the ACA, and many report that costs are the primary factor in their decision to offer health insurance or not (McGough et al., 2025; Mendoza, 2022; Long et al., 2016; Lee, 2002). Under the ACA, small and low-wage employers are able to purchase small-group health insurance plans and receive tax credits to offset premium costs, though mandated benefits mean that purchased plans remain expensive (McGough et al., 2025; Mendoza, 2022; Vistnes and Monheit, 2011).[5] Thus, strong policy and pricing mechanisms exist which continue to generate different rates of health insurance offers by firm size.

Additionally, employers are more likely to offer health insurance if their workforce is gender-balanced and younger. In 2024, approximately 60 percent of establishments with 25-74 percent female employees offered health insurance, compared to 38 percent of establishments with fewer and 44 percent of establishments with more female employees (Agency for Healthcare Research and Quality, 2024, Table I.A.2). 74 percent of establishments with 25-49 percent of workers aged 50 or older offered health insurance, compared to 23 to 51 percent of

[4] See also https://www.kff.org/faqs/faqs-health-insurance-marketplace-and-the-aca/employer-sponsored-health-coverage-and-the-marketplace/what-is-a-self-funded-self-insured-plan-how-do-i-know-if-i-have-one/.
[5] See also https://www.healthcare.gov/small-businesses/provide-shop-coverage/small-business-tax-credits/.

establishments with fewer or more older workers (Agency for Healthcare Research and Quality, 2024, Table I.A.2). Women and older workers generally consume and spend more on health care and have been more expensive to insure (Burtless, 2017; Owens, 2008; Daneshvary and Clauretie, 2007; Rappaport, 2000). Employers that self-insure (usually large firms) will take on the full costs of medical care generated by a majority female or older workforce. Since implementation of the ACA, insurers in the small-group health insurance market can no longer charge higher premiums based on gender, but can continue to differentiate premiums based on age, with limits.[6] Thus, employers choose to offer health insurance or not in part based on workforce composition and associated benefit costs.[7]

Additionally, and importantly for this analysis, the combination of firm size and worker characteristics also alters the likelihood of offering health insurance. Abraham et al. (2016) document this using MEPS-IC data while analyzing changes in health insurance offers resulting from effects of the ACA on the tax price of insurance. In establishments in small firms (49 or fewer employees), a greater share of female employees (more than 33 percent) increases the likelihood of offering health insurance, while in establishments in large firms (500 or more employees), a greater share of female employees decreases the likelihood of offering health insurance, and establishments in medium firms (50-499 employees) have similar health insurance offer rates across shares of female employees (Abraham et al., 2016). Establishments in small and medium firms with 20-50 percent of workers aged 50 or older have higher health

[6] See https://aspe.hhs.gov/reports/affordable-care-act-women-0. This is referred to as modified or adjusted community rating. Large firm premiums can still be determined by experience rating, which considers claims history, though, again, most of these organizations (76.6 percent in 2024) self-insure their health benefits (Agency for Healthcare Research and Quality Table I.A.2.a, 2024).

[7] In some industries, having a particular workforce composition may be advantageous. For example, Lee et al. (2018) find in an analysis of Korean firm productivity and demographics that having a larger share of older workers has a positive effect on productivity of manufacturing firms, particularly in large firms which are financially unstable or growing. Alternatively, employers could choose their workforce with benefit implications in mind. Employers with more generous health benefit plans are less likely to hire older workers (Scott et al., 1995).

insurance offer rates than older or younger establishments, while in establishments in large firms, the likelihood of offering health insurance declines with workforce age (Abraham et al., 2016). Thus, it is most informative to consider differences in health insurance offers by the intersection of worker characteristics and firm size. In this analysis, I use the same levels of worker characteristics as Abraham et al. (2016) to study changes in health insurance offers by the intersection of worker characteristics and firm size in the pre- and post-ACA periods.

Finally, in addition to cost and policy variation, preferences for health insurance among the workforce produce differences in health benefit offer rates. These preferences are apparent through the willingness of workers to trade off health benefits and wages. Daneshvary and Clauretie (2007) estimate earnings/employer health insurance trade-offs separately for married men and married women, using spouse's firm size and availability of family health insurance coverage from the spouse's employer as instrumental variables for receiving health insurance from one's own employer. They find that the earnings/insurance trade-off is 3.5 percentage points greater among married women than married men.[8] Cowan and Schwab (2016) conduct a difference-in-differences analysis of wage differentials between men and women that do and do not receive health insurance from their employer and find that the gender wage gap is larger for female employees who hold employer-provided health insurance than for female employees who do not receive employer-provided health insurance, which they attribute to higher expected healthcare costs for women.[9] Sheiner (1999) studies earnings/insurance trade-offs by employee age in cities with relatively high and low health costs and finds that the age/wage profile is flatter

[8] Daneshvary and Clauretie (2007) suggest that this may result from the higher expected healthcare costs of pregnancy (though they find a similar trade-off differential for women of nonreproductive ages) or because women value health insurance more highly as a result of higher average levels of risk aversion than men.
[9] This larger difference persists across age, marital status, and the presence of children, suggesting higher risk aversion as stated above.

in cities with higher health costs, such that older workers see larger decreases in wages for a given increase in health costs. Thus, women and older workers do demonstrate a higher willingness to pay for employer-provided health insurance.[10]

Worker preferences for health insurance also exist across characteristics, which employers consider when offering health insurance and workers consider when searching for and accepting a job. Bundorf (2002) finds that employers are more likely to offer health insurance if their workforce has higher average predicted health expenditures, though the effects of worker characteristics on plan choice and generosity are small. Monheit and Vistnes (1999) find evidence that workers sort themselves into jobs depending on their preference for whether or not health insurance is offered. Thus, employers will likely consider workforce preferences and composition to some extent when deciding to alter health benefits in the face of changes in costs, policy, and legislation such as minimum wages.

Heterogeneity in Minimum Wage Responses

In my analysis, I identify if differences in employer characteristics are associated with different responses to minimum wage increases. Effects may be moderated by policies which vary with firm and establishment characteristics. Marks (2011) finds that changes in health insurance offers in response to minimum wage increases vary significantly by firm size, since large firms are likely to self-insure and must offer similar health benefits to high- and low-

[10] Female and older workers' valuation of certain health benefits can vary. Gruber (1994) studies wages, hours, and employment among women of childbearing age after maternity benefits were mandated to be included in employer-provided health insurance. He finds little change in hours and employment, but wage declines on the order of 100 percent of the cost of the mandates, suggesting full valuation of the benefit. Note that Lahey (2012) finds changes in employment but no changes in wages with regard to infertility treatment mandates, concluding that infertility treatment mandates are not valued at cost. Bailey (2014) studies wages, employment, and the likelihood of having employer-provided health insurance among men aged 50-64 after health insurance plans were required to cover prostate cancer screenings. He finds that wages, employment, and the likelihood of having employer-provided health insurance fall among older men, indicating that they do not fully value the prostate cancer screening mandate (though a fall in wages suggests they value it somewhat and some are willing to pay for the mandate).

compensated employees. When minimum wages rise, workers in firms with fewer than 500 employees are less likely to report receiving employer-provided health insurance, while workers in large firms (500 or more employees) report little to no change (Marks, 2011). Similarly, the ACA provision which encourages firms with 50 or more employees to provide a minimum level of health benefits may prevent decreases in health insurance offers among these firms, if medium- to large-sized firms find that the net value of providing health insurance is greater than the employer shared responsibility payment.

Finally, and as stated previously, some employees may value health insurance more highly – thus rolling back benefits may be particularly unappealing for businesses with workers that have high anticipated health care spending. Changes in offers of health insurance in response to minimum wage increases are thus likely to vary with establishment and firm characteristics due to 1) health insurance costs; 2) impacts of the ACA and other legislation and 3) employee preferences for health insurance. Analyses so far of minimum wage effects on employer offers of health insurance generally do not consider heterogeneity by all of these features.

Data

In this analysis, I access restricted-use Agency for Healthcare Research and Quality Medical Expenditure Panel Survey - Insurance Component (MEPS-IC) data which provides information on employer features and offers of health insurance. This survey data underlies the publicly available MEPS-IC summary statistics.[11] Use of the microdata allows me to control for multiple employer features in regressions (for example, firm size, industry categorization, and

[11] For example, annual MEPS-IC summary statistics are available here: https://meps.ahrq.gov/data_stats/summ_tables/meps-ic-table-series.shtml and state-by-year summary statistics are available here: https://meps.ahrq.gov/mepsweb/data_stats/state_tables.jsp.

share of high-wage workers), account for interactions of characteristics that may affect health insurance offer decisions, and identify heterogeneity in minimum wage responses, for which there is limited evidence.

The MEPS-IC collects data from a national sample of establishments. The nationwide sample of establishments allows for comparisons in this analysis between states which increase their minimum wage above the federal level and those which do not. My analysis of minimum wage effects includes private establishments with non-missing firm size and establishment characteristics in all U.S. states, excluding Hawaii and Nevada, in 2003-2006 and 2008-2023.[12,13] I begin my analysis in survey year 2003, when sample sizes support state-level estimates for all 50 states and Washington, DC in each year.[14] The federal minimum wage increased between 2007 and 2009, so beginning in 2003 allows for analysis of health benefit offers both before and after those years. Extending my analysis up to 2023 allows for analysis of effects after implementation of the ACA and resulting from state-level minimum wage increases in the 2010s and 2020s.

The MEPS collects information on the share of low-wage, medium-wage, and high-wage workers in an establishment.[15] In 2003, low-wage workers were defined as workers receiving wages at and below $9.50 per hour. Table 1 presents the nominal wage value of a low-wage and

[12] Hawaii generally requires employers to provide health insurance to employees working 20 or more hours per week (https://cca.hawaii.gov/ins/faqs-2/hawaii-employers-faqs/). Between 2006 and 2024, Nevada had in place a two-tiered minimum wage which set different rates based on whether an employer provided health insurance (https://labor.nv.gov/About/Frequently_Asked_Questions/Frequently_Asked_Questions_-_About_Us/).

[13] MEPS data is not available for 2007, due to a change in data collection procedures. Prior to 2008, MEPS data collection occurred in the year following the reference year (so 2006 MEPS data were collected in 2007). Beginning in 2008, MEPS data collection occurred in the reference year, such that 2008 MEPS data were collected in 2008 (the time at which the 2007 data would have been collected).

[14] Prior to 2003, the 21 least populated states were not surveyed every year due to cost constraints. Beginning in 2003, all states are surveyed each year (https://meps.ahrq.gov/survey_comp/ic_technical_notes.shtml).

[15] Respondents report the share of workers receiving wages below a nominal value, at and above that value and up to a second value, and above the second value. In some cases, the share of workers within a wage category is missing, and I take the difference of 100 and the sum of the other two shares (when they are not missing and sum to 100 or less). I exclude share values outside of 0 and 100.

high-wage worker from the MEPS-IC survey instrument in each year. Table 1 also presents the highest prevailing federal or state minimum wage in the year.[16] In most cases, the definition of low-wage is greater than or equal to the highest minimum wage. However, particularly toward the end of the period, state minimum wages begin to rise faster than the low-wage definition. Thus, employers in states which raise the minimum wage are less likely to have workers in the low-wage group as the minimum wage rises. For this reason, conditioning analysis on this variable is problematic. To avoid this, in one analysis I identify effects for industries which are likely to have a large share of low-wage workers in lieu of conditioning on observed low-wage workers. In another analysis, I separate establishments into those with 50 percent or more high-wage workers and establishments with 50 percent or more low- and medium-wage workers. This allows me to identify effects among establishments for which minimum wage changes affect different shares of workers.

Additionally, the MEPS includes information on the number of employees within an establishment and within a firm. Establishment size is reported within the survey data collection, while total firm size is provided using Census Bureau Business Register data, which is used to select the MEPS sample.[17] I use firm size rather than establishment size in my analysis, for comparison to prior literature and because ACA regulations apply to firms rather than establishments. I designate firms with fewer than 50 employees small firms, those with 50-499 employees medium firms, and those with 500 or more employees large firms.[18]

[16] Minimum wage data is obtained from a dataset provided by David Neumark and from the U.S. Department of Labor Wage and Hour Division. See https://www.dol.gov/agencies/whd/state/minimum-wage/history.
[17] See https://meps.ahrq.gov/survey_comp/ic_data_collection.jsp.
[18] Note that ACA regulations actually apply to the number of full-time firm employees (https://www.irs.gov/affordable-care-act/employers/employer-shared-responsibility-provisions). The MEPS does not report full- or part-time employee counts at the firm level, so I use total employee counts to categorize firms.

The MEPS offers further information on establishment features, including the share of workers that are female or aged 50 or older.[19] Note that size categories are defined at the firm level, while characteristics are defined at the establishment level. For single-establishment firms, firm and establishment information are the same.[20]

Table 2 indicates that establishment characteristics differ slightly by firm size. The share of female employees increases with firm size, from 43 percent in establishments in small firms to 53 percent in establishments in large firms. Establishments in small firms are slightly older, with 33 percent of employees aged 50 or older on average, compared to 25 percent in medium and large firms. Interestingly, establishments in large firms have higher shares of both low-wage (below the lower value in Table 1) and high-wage (above the higher value in Table 1) workers, while establishments in small and medium firms have similar rates. Setting aside differences in health insurance offers by firm size, this pattern of characteristics itself does not predict a consistent pattern in offers of health insurance or minimum wage responses. For example, suppose that firms self-insure and fully pay benefit costs, small firms might expect lower benefit costs based on having a lower share of female workers, while medium and large firms might expect lower benefit costs based on having a lower share of older workers. A larger share of high-wage workers in large firms would imply a higher likelihood of offering health insurance, while a higher share of low-wage workers decreases that effect. Overall, establishment characteristics across firm sizes are fairly similar.

[19] I exclude cases where the share reported is less than 0 or greater than 100.
[20] As in Abraham et al. (2016), one potential limitation is that multi-establishment firms may make health insurance decisions at the firm level, rather than establishment level.

Methodology

I estimate a two-way fixed effects regression model, with the basic regression specification presented below.

$$Y_{est} = \alpha + \beta MW_{st-1} + \sigma Z_{st} + \pi X_{est} + \rho D_s + \delta D_t + \varepsilon_{est}$$

MW_{st-1} is the average prevailing federal or state minimum wage in the calendar year one year prior to the MEPS survey.[21] I use variation in the prevailing minimum wage level, induced by state legislation variation, to identify β, the effect on health insurance offers of a $1 increase in the lagged minimum wage. I expect that health insurance offers may not change contemporaneously with the minimum wage, since employers must review plan details and inform employees of changes – using a lagged minimum wage variable allows effects to occur with some delay. The regression specification includes state fixed effects D_s, year fixed effects D_t, and state-level control variables Z_{st}, including the unemployment rate, log per capita gross state product, and log per capita income from University of Kentucky Center for Poverty Research National Welfare Data. The regression specification also includes establishment-level control variables X_{est}, including industry fixed effects, firm size dummy variables (50 or more employees, 100 or more employees, and 500 or more employees), single- or multi-establishment status, and the share of workers that are female, over 50, part-time, or in a union.[22] I cluster

[21] I use the minimum wage in dollar terms to allow for comparison of my results with previous papers studying minimum wage ($) effects on health insurance offers and receipt, including Meiselbach and Abraham (2023), Dworsky et al. (2022), and Clemens et al. (2018).

[22] Economic controls account for state economic conditions which may influence employers' compensation decisions and are similar to those used in Meiselbach and Abraham (2023). Establishment controls account for other employer features over which health insurance offer rates and minimum wage responses vary. For example, Clemens (2021) notes that employers could raise prices in response to a minimum wage increase, and demand elasticities will vary between tradable and non-tradable industries. Differing (unconditional) health insurance offer rates by workforce features can be observed in the MEPS-IC annual summary statistics tables at https://meps.ahrq.gov/data_stats/summ_tables/meps-ic-table-series.shtml. I exclude cases where the share reported is less than 0 or greater than 100. I calculate the share of part-time workers in an establishment by dividing the reported number of part-time employees by the reported number of employees in the establishment.

standard errors by state to adjust for potential correlation within clusters (and because minimum wages are assigned at the state level), and weight regression estimates using MEPS final survey weights.

In heterogeneity analyses I interact the establishment characteristic variables with the minimum wage and firm size variables to identify separate minimum wage effects by the combination of characteristic and firm size. I additionally include an interaction with an indicator for the post-ACA period, which allows me to identify separate minimum wage effects by characteristic and firm size before and after the ACA. The interacted regression specification takes the form below.

$$
\begin{aligned}
Y_{est} = \alpha &+ \sum_{f=1}^{F}\sum_{j=1}^{J} \beta^{fj} MW_{st-1} \cdot 1\left(FirmSize_{est}^{f}\right) \cdot 1\left(Characteristic_{est}^{j}\right) \cdot 1(ACA_t) \\
&+ \sum_{f=1}^{F}\sum_{j=1}^{J} \gamma^{fj} 1\left(FirmSize_{est}^{f}\right) \cdot 1\left(Characteristic_{est}^{j}\right) \cdot 1(ACA_t) \\
&+ \sum_{f=1}^{F} \sigma^{fj} Z_{st} \cdot 1\left(FirmSize_{est}^{f}\right) \cdot 1(ACA_t) \\
&+ \sum_{f=1}^{F} \pi^{fj} X_{est} \cdot 1\left(FirmSize_{est}^{f}\right) \cdot 1(ACA_t) \\
&+ \sum_{f=1}^{F} \rho^{fj} D_s \cdot 1\left(FirmSize_{est}^{f}\right) \cdot 1(ACA_t) + \sum_{f=1}^{F} \delta^{fj} D_t \cdot 1\left(FirmSize_{est}^{f}\right) + \varepsilon_{est}
\end{aligned}
$$

In this specification, the one-year lagged minimum wage is interacted with an indicator for an establishment being in a small, medium, or large firm and the characteristics levels used in Abraham et al. (2016). *FirmSize* categorizes firms with fewer than 50, 50-499, and 500 or more employees. *Characteristic* (separately) categorizes the share of female workers in an

establishment into 3 categories and the share of workers aged 50 and older into 3 categories as shown in Table 3. Alternatively, *Characteristic* categorizes higher- and lower-wage establishments into two categories based on either industry or the share of workers receiving wages above a nominal value. *ACA* is an indicator variable for the pre- (2003-2013) or post- (2014-2023) ACA periods. I interact firm size and ACA indicators with all controls and fixed effects to allow for policy and firm size interactions, with separate effects by firm size and pre- and post-ACA periods. The ACA put in place different incentives to offer health insurance and opportunities to purchase health insurance by firm size, so effects may plausibly vary by the interaction of firm size and legislation. In the characteristics regressions on share female and share workers aged 50 and older, I exclude the corresponding continuous control variable in X_{est}. In most tables, I present absolute minimum wage effects by firm size or characteristic.

Results

I first identify differences in health insurance offer rates by firm size and establishment share of female and older workers, using my regression methodology excluding the minimum wage. This identifies differences by characteristics among firms in the same state, year, and industry, netting-out differences in health insurance offers which are determined by establishment features not under study. Medium and large firms are more likely to offer health insurance than small firms – prior to the ACA, medium firms offered health insurance at a rate 30 percentage points higher than small firms, increasing to 37 percentage points after implementation of the ACA (Table 3).[23] Among large firms, health insurance offers increased

[23] Note also that establishments in medium and large firms have lower shares of low-wage workers and higher shares of high-wage workers, conditional on industry and establishment features. High-wage establishments are more likely to offer health insurance, even compared to similarly-sized firms, so differences in health insurance

from 34 percentage points higher to 36 percentage points higher. A slightly larger increase among medium firms after the ACA suggests the 50-employee mandate of the ACA did bind for these firms. This conditional difference is much smaller than the observed difference in health insurance offers among firms with fewer than and 50 or more employees – the 2024 MEPS-IC summary statistics report that 32 percent of firms with fewer than 50 employees offered health insurance, compared to 96 percent of firms with 50 or more employees. Thus, state, industry, and establishment features account for a substantial portion of differences in health insurance offers across firm sizes.

In all firms, establishments with 33-66 percent female employees are the most likely to offer health insurance. The difference is much larger in establishments in small firms, which are 8 to 10 percentage points more likely to offer health insurance compared to establishments in small firms with fewer female employees. In establishments in medium and large firms, the difference is 1 percentage point. With respect to workforce age, establishments in small firms are most likely to offer health insurance in establishments with 20-50 percent older workers, and less likely to offer health insurance when the share of workers aged 50 or older is larger. Establishments in medium and large firms exhibit the same pattern, though the difference in health insurance offer rates between young and medium-aged establishments is smaller, 1 percentage point, compared to 8 percentage points among establishments in small firms. After implementation of the ACA, establishments in medium firms with fewer than 20 percent and more than 50 percent of older workers have similar health insurance offer rates, suggesting that the ACA led some establishments expected to have higher benefit costs to begin to offer health insurance. My results on establishments in small and medium firms are generally consistent with

offer rates between small and larger firms appear to be driven by both size and workforce wage composition (Agency for Healthcare Research and Quality, 2024, Table I.A.2)

those of Abraham et al., (2016), and I find small differences by characteristic category among establishments in large firms. Thus, before assessing minimum wage effects, I verify that health insurance offer rates differ by firm size-ACA interactions in the manner expected.

In Table 4, I regress health insurance offers on the minimum wage across selected samples and with and without control variables to illustrate effects of controlling for economic conditions and firm and establishment features. Columns 1 and 2 include no control variables but do include state and year fixed effects. Column 1 includes all surveyed establishments, while Column 2 includes only establishments with non-missing information on the share of female workers, workers over 50, part-time workers, low-wage workers, high-wage workers, union presence and single- or multi-establishment status (the main sample which I use in prior descriptive statistics and subsequent regressions). Columns 3 and 4 add economic controls (state unemployment rate, log per capita gross state product, and log per capita income). Columns 5 and 6 add industry fixed effects. Columns 7-11 use only the main, non-missing regression sample. Column 7 adds establishment controls, and Column 8 adds firm size dummy variables (50 or more employees, 100 or more employees, and 500 or more employees). Column 9 interacts all variables with the indicator for the post-ACA period, identifying minimum wage effects pre- and post-2014. Finally, columns 10 and 11 add a control for the share of high-wage workers in the establishment, with column 11 also including the ACA interaction.

The estimated effect of minimum wage increases on health insurance offers with or without economic controls and industry fixed effects is similar across samples. Columns 1-6 find that a $1 increase in the minimum wage yields between a 0.47 and 0.61 percentage point decrease in the share of establishments offering health insurance, with estimates close to -0.6

among the main, non-missing sample.[24] Controlling for firm size and establishment features, including the share of high-wage workers, moderates the estimate in the non-missing sample slightly to between -0.56 and -0.53. I find estimates that are similarly negative but consistently smaller than the 0.92 percentage point decline in health insurance offers found by Meiselbach and Abraham (2023), indicating that minimum wage increases do decrease health insurance offers but that using state-year summary statistics overestimates the effect. I also study effects before and after the ACA and allow effects of economic controls and firm and establishment controls to vary pre- and post-ACA, which I find changes the identified minimum wage effect. Minimum wage increases in the pre-ACA period have a negative but weak effect on health insurance offers, and the relative effect after the ACA is positive but not statistically significant. The minimum wage has virtually no effect on health insurance offers in the post-ACA period overall. I conclude that minimum wage effects on employer offers of health insurance are moderated when controlling for economic conditions and firm and establishment features, and I include these controls (excluding share high-wage) in the following regressions.

Heterogeneity in Characteristics

In Table 5, I estimate changes in health insurance offers as a result of minimum wage increases by firm size. Table 5 presents absolute effects on health insurance offers for each firm size. Offers of health insurance fall more in small firms as the minimum wage increases (0.58 percentage points with a $1 increase in the minimum wage), consistent with these firms having higher costs of providing health insurance and a greater share of low-wage workers (see Table 3). However, I conduct a regression identifying separate effects pre- and post-ACA and find that

[24] Since these estimates are reasonably similar between the full sample and the sample with non-missing employer features, I don't think excluding establishments with missing data is particularly problematic for my analysis.

effects differ in the two periods. Pre-ACA, offers of health insurance fall in small firms as the minimum wage rises, though the estimated effect is not statistically significant, with virtually no change after implementation of the ACA. Medium and large firms see no statistically significant change in health insurance offers as a result of minimum wage increases over the period, consistent with evidence from Meiselbach and Abraham (2023) and suggestive evidence from Marks (2011).

I expect that establishments with a large share of low-wage workers will experience minimum wage effects, but also that effects may be present in other establishments if these establishments still include a nonnegligible share of workers receiving wages at or near the state minimum wage. Indeed, Meiselbach and Abraham (2023) study changes in employer-provided health insurance offers by share of low-wage workers (using the MEPS low-wage definition) and find smaller but significant reductions in health insurance offers among establishments with less than 50 percent low-wage workers. In Table 6, I estimate changes by firm size and two measures of low-wage worker concentration.

In the first set of regressions, in columns 1-3, I interact the minimum wage with firm size variables and a dummy variable for an establishment being in the retail or accommodation and food service industries. Restaurant and retail industries are commonly used in minimum wage analyses to identify large shares of likely low-wage workers who may be subject to minimum wage effects (Neumark and Wohl, 2026; Reich et al., 2017; Dube et al., 2010; Sabia, 2009; Card and Krueger, 1994). I find that, over the full period, a $1 minimum wage increase leads to a 0.5 percentage point decline in health insurance offers in small, likely low-wage establishments and a 0.6 percentage point decrease in health insurance offers in other small establishments, again consistent with small firms being most responsive to wage and benefit cost increases. In the pre-

ACA period, small, likely low-wage establishments decrease offers of health insurance by 1.5 percentage points as the minimum wage increases $1, with no effect in the post-ACA period. In medium, likely low-wage establishments, offers of health insurance increase in response to minimum wage increase in the pre-ACA period. This is particularly surprising – while medium-sized firms are more likely to offer health insurance than smaller firms, these effects occur prior to the ACA employer mandate, and these firms are less likely than larger firms to self-insure and be subject to non-discrimination requirements.

In the second set of regressions, in columns 4-6, I use share of high-wage workers to split firms and establishments. High-wage establishments are those with 50 percent or more of workers receiving wages at or above the higher value identified in the MEPS-IC survey (and listed in Table 1). In the non-high-wage establishments, at least 50 percent of workers receive wages below that wage value (such that 50 percent of workers are low- or medium-wage). Meiselbach and Abraham (2023) report estimates in businesses based on the share of workers receiving wages below the low-wage value listed in Table 1. However, this is problematic because the MEPS-IC low-wage definition is lower than some state minimum wages, particularly in the end of period, such that establishments in these states could be classified as not low-wage when they are paying minimum wages. As a result, minimum wage effects, especially from increases towards the end of the period, could be attributed to "high-wage" establishments when the effects are actually occurring among (relative) low-wage establishments within a state. My approach, which uses the higher of the nominal wage values, avoids direct impacts of the minimum wage on categorization of establishments. Over the period, I find again the largest reductions in health insurance offers from minimum wage increases occur among establishments in small firms, both high-wage and not high-wage, with a decline in health

insurance offers of approximately 0.6 percentage points with a $1 increase in the minimum wage. Offers of health insurance weakly decrease as the minimum wage rises in small, not high-wage businesses prior to ACA implementation. After ACA implementation, small, not high-wage establishments might again be expected to reduce health insurance offers, since these establishments are not subject to the employer mandate of the ACA and minimum wage increases will continue to increase labor costs. Instead, I find a near-zero effect among these firms in the post-ACA period. It is possible that introduction of the small-group market allowed small-firm establishments to continue to offer health insurance in the post-ACA period, even as minimum wages rose. It is clearer, in comparing columns 4-6, that the inclusion of ACA timing interaction with control variables moderated estimated minimum wage effects – in column 4, which includes the entire period with no ACA interactions, small-firm, not high-wage establishments see a significant decrease in health insurance offers with minimum wage increases, while effects are negative and not statistically significant in column 5 and very close to zero in column 6, with the latter two columns including ACA-control interactions.

Although I use a different variable to identify high-wage establishments than Meiselbach and Abraham (2023), I similarly find reductions in health insurance offers among higher-wage establishments as the minimum wage increases. This suggests that the health insurance effect of minimum wages spills over to establishments where the minimum wage is less binding. For example, businesses may decide to alter benefits even if only a small share of workers receive the minimum wage due to benefit costs and equity, or if minimum wage increases lead similarly-situated establishments within the labor market to reduce benefits. Overall, I identify effects on health insurance offers across wage categories and conclude that reductions in health insurance offers as a result of minimum wage increases occur in establishments in small firms.

In Table 7, I study differences in minimum wage effects on health insurance offers by firm size and establishment share of female and older workers. In the following regressions, I present only results for the pre- and post-ACA periods, rather than the full sample period, because economic control and firm and establishment control effects on health insurance offers seem to vary significantly – and thus affect estimated minimum wage effects – in the period before and after ACA implementation.

In small firms, minimum wage increases lead to a larger decline in health insurance offers in establishments with fewer female employees (Panel A). Pre-ACA, a $1 increase in the minimum wage decreases health insurance offers 0.8 percentage points in establishments with less than 33 percent female employees and 0.2 percentage points in establishments with more than 66 percent female employees (with the relative difference significant at the 10 percent level). After implementation of the ACA, health insurance offers decline 0.4 percentage points in establishments with few female workers and actually increase 0.5 percentage points in establishments in small firms with a large share of female workers. An increase in health insurance offers after a minimum wage increase is surprising, although the ACA's elimination of sex in premium rating may have made covering a more-female workforce cheaper for establishments in small firms in this period relative to pre-ACA. Larger declines among establishments with fewer female employees is consistent with female employees having higher expected medical costs and valuing health insurance more highly. In establishments with a greater share of female workers, reductions in health insurance offers will be more undesirable for the workforce as a whole than in establishments with fewer female employees. I find a similar though weaker pattern among establishments in medium and large firms in the pre-ACA period. Following implementation of the ACA, health insurance offers among establishments in

medium firms similarly decline most in those with the fewest female employees. Offers of health insurance are unchanged in large firms post-ACA.

With respect to workforce age, I find that a $1 increase in the minimum wage reduces health insurance offers 1 percentage point in small-firm establishments with less than 20 percent of workers aged 50 or older in the pre-ACA period (Panel B). This is the largest decline in small firms prior to the ACA, consistent with these workers valuing health insurance less than older workers and possibly being more willing to bear benefit declines. In the post-ACA period, small-firm establishments with moderate shares of older workers see an increase in health insurance offers of 0.6 percentage points with a $1 increase in the minimum wage, while establishments with fewer or more older workers see almost no change in health insurance offers. The decline in health insurance offers in the post-ACA period is smaller than the pre-ACA period, indicating that the ACA moderates reductions, but the pre-/post- difference with respect to age is less than with respect to sex. Notably, in the post-ACA period age can still be considered in small-firm premium calculation in a way that sex cannot, such that small-firm premiums for older workers likely do not decline in the same way as for female workers. In establishments in medium and large firms, health insurance offers increase somewhat with the minimum wage for establishments with more than 50 percent older workers. Again, benefit and wage increases simultaneously are surprising, but these establishments have workforce characteristics such that employees may value health insurance very highly and would not desire its removal.[25]

[25] In Appendix Table A1, I further interact establishment characteristics with the high-wage variable used in Table 6 to identify effects in lower- and higher-wage establishments. I interact all variables with the high-wage indicator to identify separate effects in high-wage and not-high-wage establishments. High-wage establishments in small firms exhibit responses in health insurance offers to minimum wage increases, particularly before ACA implementation. Among high-wage establishments in small firms, declines in health insurance offers are largest in establishments that are predominately male or female, with declines of 2 to 3 percentage points with a $1 increase in the minimum wage. After ACA implementation, both lower- and higher-wage small establishments with a large share of female employees see increases in health insurance offers (0.5 to 1 percentage point) with the minimum wage, consistent with these workforces having the highest valuation of health insurance. With respect to age, high-wage small

Composition

Finally, I test if the share of high-wage, female, or older workers changes in establishments as the minimum wage increases and before and after ACA implementation. If workforce characteristics change with the minimum wage, compositional effects may contribute to the effects I identify. For example, if employers choose to hire fewer or more female employees with the minimum wage, they may also change their health insurance offers in a manner associated with changes in the share of female workers.

In Table 8 I find no significant effects of minimum wage changes on the share of female workers or workers aged 50 and older in establishments. I find that the share of high-wage workers in establishments in large firms increases slightly, 0.3 percentage points with a $1 increase in the minimum wage, but no other effects, and conclude that the health insurance offer changes I identify do not appear to stem from compositional changes in the sample.[26]

establishments do reduce health insurance offers in response to minimum wage increases in the pre-ACA period, with the largest reduction among establishments with fewer than 20 percent older workers. This minimum wage effect appears to dissipate in the post-ACA period, which may result from the introduction of modified community ratings limiting premium costs with respect to age. Again, I find reductions in health insurance offers largely among small firms, though the presence of effects in both lower- and higher-wage small businesses is surprising. Higher-wage businesses are more likely to offer health insurance, even among small employers, though health insurance offer rates are low – in 2024, 37 percent of establishments in small firms with fewer than 50 percent low-wage employees offered health insurance, compared to 17 percent of establishments in small firms with more low-wage workers (Agency for Healthcare Research and Quality, 2024, Table I.A.2). However, as Clemens (2021) points out, approximately half of establishments that offer health insurance offer only one plan, such that effects on a common benefits plan can spill over to both minimum wage and non-minimum wage employees. The rate is even larger among establishments in small firms, with approximately 65 percent of businesses that offer a plan offering only one health benefit plan (Agency for Healthcare Research and Quality, 2024, Table I.A.2.d). The presence of effects in both sets of small firms, which have higher benefit costs, suggests that the interaction of benefit costs and minimum wage costs may be salient, even if the minimum wage is less binding among an employer's workforce.

[26] Note that there is emerging evidence that minimum wage increases cause reallocation of low-wage workers from small establishments to larger (Dustmann et al., 2022). I don't find evidence of compositional changes within the workforce by firm size. However, my analysis does implicitly ignore firm exit, which could be associated with reallocation. Null changes in the share of female and older workers by firm size suggest that employees with a higher valuation of health insurance are not reallocating to larger firms.

Conclusion

In this analysis, I use MEPS-IC microdata to study changes in employer health insurance offers after minimum wage increases. This dataset allows me to answer the first-order question of how employers adjust health benefit offers in response to minimum wage increases and identify heterogeneous effects which result from differences in benefit costs, policy, and employee preferences for insurance.

Minimum wage effects on health insurance offers are driven by changes in small firms, with 49 or fewer employees, which face higher costs of providing health insurance than larger firms. I find reductions in health insurance offers as the minimum wage increases in both low-wage and high-wage establishments in small firms, suggesting that benefit costs are particularly salient, even in businesses where minimum wage increases are less binding. Health insurance offer reductions are smaller in establishments where the workforce likely values employer-provided health insurance highly (namely, establishments with a greater share of female or older workers). Policies within the ACA which limited premium costs based on workforce sex and age may have moderated declines in health insurance offers among the same establishments.

This analysis characterizes one area in which employers alter labor practices in response to minimum wage increases, adding to literature on changes in employment, hours, training, and benefits. Future areas for analysis include other aspects of health insurance plans, such as premiums and dependent coverage, which affect workers' decision to take-up health insurance. Estimates of changes in health insurance plan features will enable further comparison of employers' changes in health insurance offers and benefits and employees' take-up of benefits, identifying for whom minimum wages are most likely to change sources and costs of health insurance.

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Tables

Table 1. MEPS-IC Low-Wage, High-Wage Definitions and State Minimum Wage by Year

Year	Low-Wage (MEPS Definition)	High-Wage (MEPS Definition)	Highest State Minimum Wage
2003	9.50	21.00	7.15
2004	10.00	22.50	7.16
2005	10.00	23.00	7.35
2006	10.50	23.50	7.63
2008	11.00	25.50	8.07
2009	11.00	25.50	8.55
2010	11.50	26.00	8.55
2011	11.50	26.50	8.67
2012	11.50	27.00	9.05
2013	11.50	27.00	9.19
2014	11.50	27.50	9.50
2015	11.50	28.00	10.50
2016	11.50	28.50	11.50
2017	12.00	29.50	12.50
2018	12.00	30.00	13.25
2019	12.50	30.50	14.00
2020	13.00	31.00	15.00
2021	13.50	32.00	15.20
2022	14.50	34.00	16.10
2023	15.50	34.00	17.00

"Low-wage" workers receive wages below the lower value, high-wage workers receive wages above the higher value, and medium-wage workers receiving wages between the two levels. See https://meps.ahrq.gov/mepsweb/survey_comp/survey_ic.jsp for MEPS-IC questionnaires by year. Minimum wage data obtained from dataset provided by David Neumark and https://www.dol.gov/agencies/whd/state/minimum-wage/history.

Table 2. Establishment Characteristics by Firm Size

	(1) Small firms	(2) Medium firms	(3) Large firms
Share female	43.48	50.80	53.06
	(0.07)	(0.12)	(0.10)
Share 50 or older	33.55	25.32	25.37
	(0.07)	(0.09)	(0.07)
Share low-wage	27.59	27.55	30.36
	(0.07)	(0.13)	(0.13)
Share high-wage	18.66	18.66	23.80
	(0.06)	(0.10)	(0.10)
N	268,000	68,000	64,500

MEPS-IC data, 2003-2023. Sample includes all establishments with non-missing characteristics (share female, share 50 or older, share part-time, share in a union, share low-wage, share high-wage, single or multi-establishment), excluding those in Hawaii and Nevada. Small firms have 49 or fewer employees, medium firms have 50-499 employees, and large firms have 500 or more employees. Shares are measured on a scale from 0 to 100. Standard errors are reported in parentheses. Weighted using MEPS final weights.

Table 3. Shares of Low-Wage, High-Wage Workers and Health Insurance Offers by Firm Size and Establishment Characteristics

	Share Low-Wage		Share High-Wage		Share Offering Health Insurance	
	(1)	(2)	(3)	(4)	(5)	(6)
	Pre-ACA	Post-ACA	Pre-ACA	Post-ACA	Pre-ACA	Post-ACA
Panel A						
Small firms	Ref	Ref	Ref	Ref	Ref	Ref
Medium firms	-0.0015	-0.0155**	0.0196***	0.0100**	0.3008***	0.3665***
	(0.0044)	(0.0060)	(0.0026)	(0.0044)	(0.0071)	(0.0088)
Large firms	-0.0176***	-0.0270***	0.0667***	0.0781***	0.3350***	0.3623***
	(0.0058)	(0.0079)	(0.0059)	(0.0049)	(0.0102)	(0.0090)
N	400,000		400,000		400,000	

Table 4. Shares of Low-Wage, High-Wage Workers and Health Insurance Offers by Firm Size and Establishment Characteristics (cont.)

	Share Low-Wage		Share High-Wage		Share Offering Health Insurance	
	(1)	(2)	(3)	(4)	(5)	(6)
	Pre-ACA	Post-ACA	Pre-ACA	Post-ACA	Pre-ACA	Post-ACA
Panel B						
Small firms						
<33% female	Ref	Ref	Ref	Ref	Ref	Ref
33-66% female	0.0352***	0.0133**	-0.0143***	-0.0237***	0.1020***	0.0801***
	(0.0042)	(0.0052)	(0.0029)	(0.0058)	(0.0080)	(0.0105)
>66% female	0.0819***	0.0506***	-0.0818***	-0.0754***	0.0136	-0.0044
	(0.0073)	(0.0077)	(0.0031)	(0.0064)	(0.0082)	(0.0087)
Medium firms						
<33% female	Ref	Ref	Ref	Ref	Ref	Ref
33-66% female	0.0707***	0.0590***	-0.0034	0.0047	0.0023	0.0072
	(0.0060)	(0.0065)	(0.0045)	(0.0058)	(0.0084)	(0.0076)
>66% female	0.1189***	0.0852***	-0.0649***	-0.0538***	-0.0239**	-0.0216***
	(0.0080)	(0.0103)	(0.0079)	(0.0080)	(0.0108)	(0.0079)
Large firms						
<33% female	Ref	Ref	Ref	Ref	Ref	Ref
33-66% female	0.0243***	0.0590***	0.0127*	-0.005	0.0104**	0.0092***
	(0.0054)	(0.0065)	(0.0064)	(0.0063)	(0.0041)	(0.0029)
>66% female	0.0931***	0.0852***	-0.0637***	-0.0718***	-0.0012	-0.0005
	(0.0081)	(0.0103)	(0.0063)	(0.0089)	(0.0053)	(0.0045)
N	400,000		400,000		400,000	

Table 5. Shares of Low-Wage, High-Wage Workers and Health Insurance Offers by Firm Size and Establishment Characteristics (cont.)

	Share Low-Wage		Share High-Wage		Share Offering Health Insurance	
	(1)	(2)	(3)	(4)	(5)	(6)
	Pre-ACA	Post-ACA	Pre-ACA	Post-ACA	Pre-ACA	Post-ACA
Panel C						
Small firms						
<20% 50 or older	Ref	Ref	Ref	Ref	Ref	Ref
20-50% 50 or older	-0.0409***	-0.0421***	0.0211***	0.0113***	0.0842***	0.0819***
	(0.0027)	(0.0053)	(0.0026)	(0.0032)	(0.0051)	(0.0074)
>50% 50 or older	-0.0086**	0.0025	0.0531***	0.0485***	-0.0516***	-0.0542***
	(0.0041)	(0.0058)	(0.0045)	(0.0066)	(0.0047)	(0.0067)
Medium firms						
<20% 50 or older	Ref	Ref	Ref	Ref	Ref	Ref
20-50% 50 or older	-0.0526***	-0.0307***	0.0431***	0.0476***	0.0147***	0.0104***
	(0.0072)	(0.0046)	(0.0032)	(0.0041)	(0.0044)	(0.0029)
>50% 50 or older	-0.0883***	-0.0432***	0.0742***	0.0709***	-0.0163***	-0.004
	(0.0078)	(0.0063)	(0.0088)	(0.0149)	(0.0058)	(0.0053)
Large firms						
<20% 50 or older	Ref	Ref	Ref	Ref	Ref	Ref
20-50% 50 or older	-0.0664***	-0.0683***	0.0873***	0.0868***	0.0114***	0.0055*
	(0.0062)	(0.0093)	(0.0042)	(0.0073)	(0.0021)	(0.0028)
>50% 50 or older	-0.0890***	-0.0875***	0.0860***	0.0941***	-0.0102*	-0.0114
	(0.0105)	(0.0101)	(0.0111)	(0.0115)	(0.0056)	(0.0076)
N	400,000		400,000		400,000	

MEPS 2003-2023 data, excluding establishments in Nevada and Hawaii and any with missing characteristics (share female, share 50 or older, share part-time, share in a union, share low-wage, share high-wage, single or multi-establishment). Reference categories indicated. Small firms have 49 or fewer employees, medium firms have 50-499 employees, and large firms have 500 or more employees. Control variables include industry, percent share female, percent share over the age of 50, percent share working part-time, presence of a union (indicator if share of workers in a union greater than 0), single- or multi-establishment, and firm size control (if employment is greater than or equal to 100 and if employment is greater than or equal to 500). In Panel B, share female is dropped, and in Panel C, share over the age of 50 is dropped. State- and year-fixed effects included, and standard errors clustered by state. Weighted using MEPS final weights.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 6. Whole Sample Regression of Health Insurance Offers, Adding Controls and Altering Sample

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)
MW_{t-1}	-0.0051**	-0.0061**	-0.0049**	-0.0059**	-0.0047**	-0.0058**	-0.0056**	-0.0053**	-0.0025	-0.0054**	-0.0023
	(0.0023)	(0.0029)	(0.0020)	(0.0025)	(0.0018)	(0.0024)	(0.0026)	(0.0025)	(0.0045)	(0.0024)	(0.0044)
MW_{t-1} X ACA_t									0.0028		0.0024
									(0.0053)		(0.0051)
Economic controls			Y	Y	Y	Y	Y	Y	Y	Y	Y
Industry controls					Y	Y	Y	Y	Y	Y	Y
Characteristic controls							Y	Y	Y	Y	Y
Firm size controls								Y	Y	Y	Y
High-wage control										Y	Y
N	538,000	400,000	538,000	400,000	538,000	400,000	400,000	400,000	400,000	400,000	400,000

MEPS 2003-2023 data, excluding establishments in Nevada and Hawaii. Columns 1, 3, and 5 include all establishments, and Columns 2, 4, and 6-11 includes only establishments with non-missing characteristics (share female, share 50 or older, share part-time, share in a union, share low-wage, share high-wage, single or multi-establishment). Control variables are added to regressions in the order presented, and include state economic conditions (log gross state product per capita, log personal income per capita, and state unemployment rate), industry division fixed effects, establishment characteristics (percent share female, percent share over the age of 50, percent share working part-time, presence of a union (indicator if share of workers in a union greater than 0), and single- or multi-establishment), firm size control (if employment is greater than or equal to 100 and if employment is greater than or equal to 500), and the share of workers in the establishment reported to receive wages above the value defined by the MEPS-IC questionnaire. In all regressions, state- and year-fixed effects included, and standard errors clustered by state. Weighted using MEPS final weights.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 7. Regression of Health Insurance Offers by Firm Size

	(1) Entire Period	(2) Pre-ACA	(3) Post-ACA
MW_{t-1}·Small Firms	-0.0058**	-0.0053	0.0008
	(0.0028)	(0.0053)	(0.0021)
MW_{t-1}·Medium Firms	-0.0011	0.0091	-0.0017
	(0.0024)	(0.0060)	(0.0026)
MW_{t-1}·Large Firms	-0.0006	-0.0035	0.0007
	(0.0007)	(0.0063)	(0.0010)
N	400,000	400,000	

See Table 3 notes. Columns 2 and 3 present estimates from a single regression.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 8. Regression of Health Insurance Offers by Firm Size, Likely Low-Wage Industries, and Share High-Wage

	(1) Entire Period	(2) Pre-ACA	(3) Post-ACA	(4) Entire Period	(5) Pre-ACA	(6) Post-ACA
MW_{t-1}·Small, Likely Low-Wage	-0.0045*	-0.0154*	0.0004			
	(0.0025)	(0.0086)	(0.0030)			
MW_{t-1}·Medium, Likely Low-Wage	0.0026	0.0146*	-0.00003			
	(0.0025)	(0.0076)	(0.0033)			
MW_{t-1}·Large, Likely Low-Wage	-0.0003	-0.0026	0.0006			
	(0.0007)	(0.0057)	(0.0012)			
MW_{t-1}·Small, Other	-0.0060*	-0.0031	0.0008			
	(0.0031)	(0.0048)	(0.0025)			
MW_{t-1}·Medium, Other	-0.0022	0.0075	-0.0021			
	(0.0026)	(0.0062)	(0.0027)			
MW_{t-1}·Large, Other	-0.0006	-0.0040	0.0007			
	(0.0008)	(0.0067)	(0.0011)			
MW_{t-1}·Small, Not High-Wage				-0.0056**	-0.0060	0.0003
				(0.0026)	(0.0060)	(0.0022)
MW_{t-1}·Medium, Not High-Wage				-0.0009	0.0096	-0.0017
				(0.0022)	(0.0060)	(0.0026)
MW_{t-1}·Large, Not High-Wage				-0.0002	-0.0022	0.0005
				(0.0008)	(0.0066)	(0.0011)
MW_{t-1}·Small, High-Wage				-0.0066*	-0.0008	0.0016
				(0.0035)	(0.0058)	(0.0028)
MW_{t-1}· Medium, High-Wage				-0.0019	0.0044	-0.0017
				(0.0032)	(0.0078)	(0.0030)
MW_{t-1}·Large, High-Wage				-0.0015**	-0.0082	0.0011
				(0.0006)	(0.0057)	(0.0009)
N	400,000	400,000		400,000	400,000	

See Table 3 notes. "Likely low-wage" firms/establishments are those in the retail and accommodation/food industries. "High-wage" firms/establishments are those with 50 percent or more of workers in the establishment reported to receive wages above the value defined by the MEPS-IC questionnaire.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 9. Heterogeneity in Health Insurance Offer Minimum Wage Effects by Firm Size and Establishment Characteristics

	(1) Small firms	(2) Medium firms	(3) Large firms	N
Panel A				
Pre-ACA				
MW_{t-1}·<33% female	-0.0079	0.0043	-0.0097	
	(0.0063)	(0.0071)	(0.0066)	
MW_{t-1}·33-66% female	-0.0067	0.0091	-0.0033	
	(0.0047)	(0.0078)	(0.0059)	
MW_{t-1}·>66% female	-0.0016	0.0133**	0.0010	
	(0.0054)	(0.0060)	(0.0075)	
				400,000
Post-ACA				
MW_{t-1}·<33% female	-0.0041*	-0.0042*	0.00004	
	(0.0024)	(0.0024)	(0.0011)	
MW_{t-1}·33-66% female	0.0034	0.0007	0.0015	
	(0.0047)	(0.0034)	(0.0011)	
MW_{t-1}·>66% female	0.0051**	-0.0016	0.00001	
	(0.0021)	(0.0030)	(0.0016)	
Panel B				
Pre-ACA				
MW_{t-1}·<20% 50 or older	-0.0102*	0.0070	-0.0020	
	(0.0054)	(0.0064)	(0.0067)	
MW_{t-1}·20-50% 50 or older	0.0021	0.0094	-0.0048	
	(0.0049)	(0.0070)	(0.0060)	
MW_{t-1}·>50% 50 or older	-0.0037	0.0154**	0.0023	
	(0.0064)	(0.0070)	(0.0094)	
				400,000
Post-ACA				
MW_{t-1}·<20% 50 or older	-0.0014	-0.0019	-0.0012	
	(0.0026)	(0.0027)	(0.0014)	
MW_{t-1}·20-50% 50 or older	0.0063***	-0.0025	0.0009	
	(0.0021)	(0.0026)	(0.0010)	
MW_{t-1}·>50% 50 or older	-0.0007	0.0012	0.0050**	
	(0.0028)	(0.0028)	(0.0020)	

See Table 3 notes. Each panel presents results from a separate regression. In Panel A, share female is dropped, and in Panel B, share over the age of 50 is dropped.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 10. Regression of Changes in Establishment Characteristics

	(1) Share High-Wage	(2) Share Female	(3) Share 50 or Older
Pre-ACA			
MW_{t-1}·Small Firms	-0.0014	0.0014	0.0018
	(0.0026)	(0.0034)	(0.0052)
MW_{t-1}·Medium Firms	0.0024	0.0020	0.0048
	(0.0061)	(0.0074)	(0.0050)
MW_{t-1}·Large Firms	-0.0036	-0.0051	0.0029
	(0.0092)	(0.0108)	(0.0069)
Post-ACA			
MW_{t-1}·Small Firms	0.0005	-0.0011	0.0025
	(0.0016)	(0.0018)	(0.0022)
MW_{t-1}·Medium Firms	0.0026	0.0036	0.0026
	(0.0021)	(0.0026)	(0.0033)
MW_{t-1}·Large Firms	0.0038*	0.0003	0.0019
	(0.0019)	(0.0022)	(0.0025)
N	400,000	400,000	400,000

See Table 3 notes. Shares are measured on a scale from 0 to 1. In each regression, the outcome variable is excluded as a control variable.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Appendix Tables

Appendix Table A1. Heterogeneity in Health Insurance Offer Minimum Wage Effects by Firm Size, Wage Level, and Establishment Characteristics

	(1) Small firms	(2) Medium firms	(3) Large firms	N
Panel A				
Pre-ACA				
MW_{t-1}·Not High-Wage·<33% female	-0.0019	0.0045	-0.0120	
	(0.0063)	(0.0070)	(0.0080)	
MW_{t-1}·Not High-Wage·33-66% female	-0.0057	0.0089	-0.0031	
	(0.0050)	(0.0079)	(0.0073)	
MW_{t-1}·Not High-Wage·>66% female	0.0046	0.0133**	0.0006	
	(0.0049)	(0.0060)	(0.0088)	
MW_{t-1}·High-Wage·<33% female	-0.0340***	-0.0001	0.0021	
	(0.0120)	(0.0160)	(0.0028)	
MW_{t-1}·High-Wage·33-66% female	-0.0151	0.0042	0.0002	
	(0.0137)	(0.0125)	(0.0022)	
MW_{t-1}·High-Wage·>66% female	-0.0253*	0.0069	-0.0092	
	(0.0151)	(0.0133)	(0.0064)	
				400,000
Post-ACA				
MW_{t-1}·Not High-Wage·<33% female	-0.0026	-0.0060**	-0.0001	
	(0.0027)	(0.0024)	(0.0017)	
MW_{t-1}·Not High-Wage·33-66% female	0.0037	0.0019	0.0016	
	(0.0049)	(0.0038)	(0.0014)	
MW_{t-1}·Not High-Wage·>66% female	0.0054*	-0.0032	-0.0001	
	(0.0028)	(0.0029)	(0.0019)	
MW_{t-1}·High-Wage·<33% female	-0.0093**	0.0072	0.0010	
	(0.0038)	(0.0047)	(0.0014)	
MW_{t-1}·High-Wage·33-66% female	-0.0025	0.0035	0.0006	
	(0.0057)	(0.0053)	(0.0009)	
MW_{t-1}·High-Wage·>66% female	0.0105*	0.0056	0.0024*	
	(0.0056)	(0.0047)	(0.0013)	

Appendix Table A1. Heterogeneity in Health Insurance Offer Minimum Wage Effects by Firm Size, Wage Level, and Establishment Characteristics (cont.)

	(1) Small firms	(2) Medium firms	(3) Large firms	N
Panel B				
Pre-ACA				
MW_{t-1}·Not High-Wage·<20% 50 or older	-0.0041	0.0074	-0.0031	
	(0.0055)	(0.0064)	(0.0079)	
MW_{t-1}·Not High-Wage·20-50% 50 or older	0.0060	0.0089	-0.0052	
	(0.0051)	(0.0063)	(0.0075)	
MW_{t-1}·Not High-Wage·>50% 50 or older	0.0010	0.0210***	0.0032	
	(0.0063)	(0.0077)	(0.0123)	
MW_{t-1}·High-Wage·<20% 50 or older	-0.0318***	0.0034	-0.0026	
	(0.0118)	(0.0139)	(0.0046)	
MW_{t-1}·High-Wage·20-50% 50 or older	-0.0193	-0.0011	0.0009	
	(0.0132)	(0.0112)	(0.0023)	
MW_{t-1}·High-Wage·>50% 50 or older	-0.0251	0.0103	0.0057	
	(0.0150)	(0.0164)	(0.0051)	400,000
Post-ACA				
MW_{t-1}·Not High-Wage·<20% 50 or older	-0.0017	-0.0024	-0.0016	
	(0.0031)	(0.0028)	(0.0016)	
MW_{t-1}·Not High-Wage·20-50% 50 or older	0.0064**	-0.0033	0.001	
	(0.0025)	(0.0025)	(0.0013)	
MW_{t-1}·Not High-Wage·>50% 50 or older	0.0015	0.0007	0.0051**	
	(0.0030)	(0.0027)	(0.0024)	
MW_{t-1}·High-Wage·<20% 50 or older	-0.0003	0.0053	0.0020*	
	(0.0042)	(0.0049)	(0.0010)	
MW_{t-1}·High-Wage·20-50% 50 or older	0.0030	0.0041	0.0007	
	(0.0054)	(0.0052)	(0.0011)	
MW_{t-1}·High-Wage·>50% 50 or older	-0.0073	0.0086	0.0036**	
	(0.0051)	(0.0057)	(0.0018)	

See Table 3 notes. Each panel presents results from a separate regression. In Panel A, share female is dropped, and in Panel B, share over the age of 50 is dropped. "High-wage" firms/establishments are those with 50 percent or more of workers in the establishment reported to receive wages above the value defined by the MEPS-IC questionnaire. "High-wage" is interacted with all variables in the regression.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$